

B. Tech 1st Semester 2nd Sessional Examination 2024 (Sec C, CE)
Subject: Mathematics-I **Subject Code: BMA23111T**

Marks-30

1. If $z = f(x + ay) + \varphi(x - ay)$, prove that $\frac{\partial^2 z}{\partial y^2} = a^2 \left(\frac{\partial^2 z}{\partial x^2} \right)$ (6)
2. Verify the Euler's theorem of the function $u = \frac{x(x^3 - y^3)}{x^3 + y^3}$. (6)
3. Evaluate $\int_{-a}^a x \sqrt{a^2 - x^2} dx$ (6)
4. Test the convergence of following series (6)
$$\sum_{n=1}^{\infty} \frac{n^2 + 1}{n^4 + 2n}$$
5. Test the convergence of the series $\sum_{n=1}^{\infty} \left[\frac{1}{n^2} + \frac{1}{(n+1)^2} \right]$ (6)